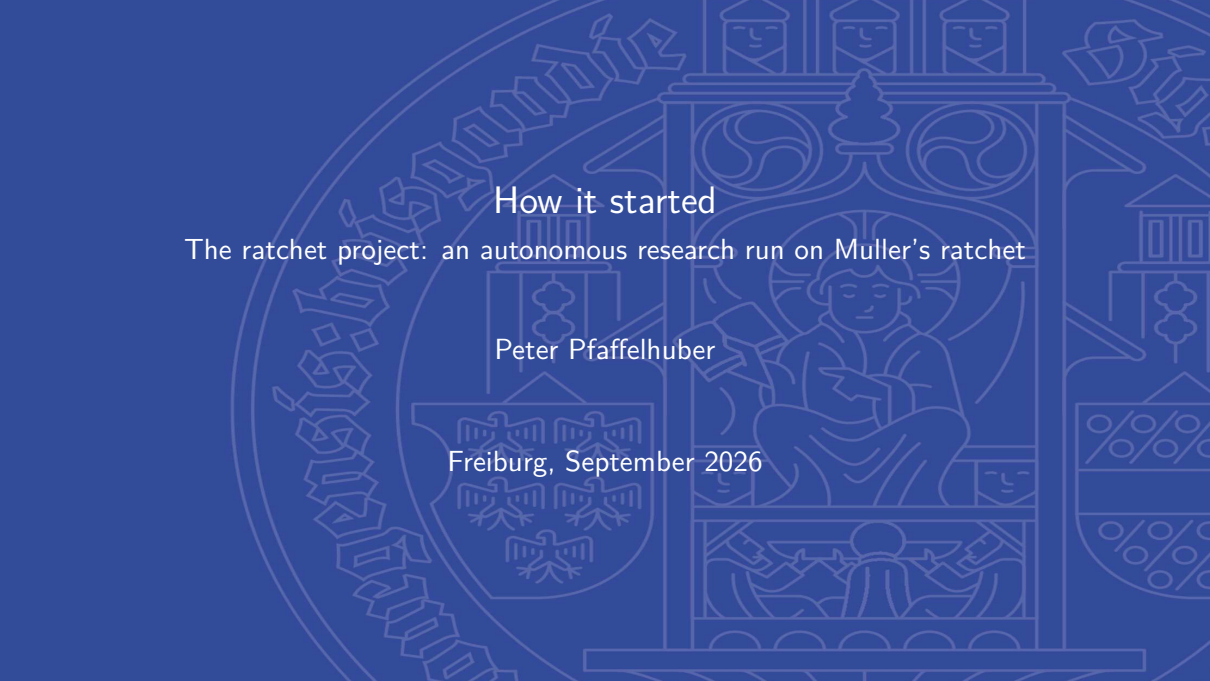


The background of the slide features a large, faint watermark of the University of Freiburg seal. The seal is circular and contains a central figure, likely a seated person, surrounded by various heraldic symbols and Latin text. The watermark is rendered in a light blue color against a darker blue background.

How it started

The ratchet project: an autonomous research run on Muller's ratchet

Peter Pfaffelhuber

Freiburg, September 2026

The problem: the click rate of Muller's ratchet

An infinite system of SDEs with Fleming–Viot noise, $k = 0, 1, 2, \dots$, $\sum_k X_k = 1$:

$$dX_k = \left(\alpha \left(\sum_j (j - k) X_j \right) X_k + \lambda (X_{k-1} - X_k) \right) dt + \sum_{\ell \neq k} \sqrt{\frac{1}{\nu} X_k X_\ell} dW_{k\ell}.$$

X_k is the fraction of individuals carrying k deleterious mutations. $K^*(t) = \inf\{k : X_k(t) > 0\}$ occasionally jumps by one: a **click**.

- ▶ Wanted: the rate $\mathfrak{r} = \lim_t K^*(t)/t$. Since $\mathfrak{r}(\alpha, \lambda, \nu) = \alpha \mathfrak{r}(1, \theta, \nu\alpha)$, there are two parameters: $\theta = \lambda/\alpha$ and $n = \nu\alpha$.
- ▶ For $n \rightarrow \infty$ this is **small noise**. The fixed point is $\pi = \text{Poi}(\theta)$, and a click is a rare exit. Conjecture: $\frac{1}{n} \log \mathfrak{r} \rightarrow -V_*(\theta)$, with V_* the quasi-potential.
- ▶ The literature closes the system into one equation for X_0 : $V_{1d}(\theta) = 2\theta[(1 + \theta) \log(1 + 1/\theta) - 1]$. The error is up to a factor of 2.

The task

Get beyond the 1d heuristic. Another 1d closure is **not** progress.

What was available

In the repository

- ▶ our own manuscript arXiv:2606.15842 (Heinzel, P., Wakolbinger), PDF and T_EX
- ▶ `ldp.tex`, “state of the discussion”: the Hamiltonian, the Neher–Shraiman heuristic, two checklists
- ▶ 93 ground-truth files: $N = 10^4$, 1000 paths each up to the first click
- ▶ the Wright–Fisher simulation from the paper, CPU port and GPU

Local, not in the repository

(~/bwSyncShare/Papers)

- ▶ Feng & Kurtz, *Large Deviations for Stochastic Processes*, 414 pp.
- ▶ Feng & Xiong (2002): quasi-potential for Fleming–Viot, “the blueprint”

Ratchet literature (Papers/ratchet/)

- ▶ Neher & Shraiman (2012): the 1d heuristic
- ▶ Audiffren & Pardoux (2013); Mariani, Pardoux & Velleret (2025)
- ▶ Etheridge, P., Wakolbinger (2008)
- ▶ Dawson & Feng (1998): LDP for Fleming–Viot *with selection*
- ▶ Champagnat & Hass (2023) and seven more

Obtained on 13 Aug for B4

- ▶ Conforti–Kraaij–Tonon (2), Puhalskii (2), Budhiraja–Dupuis–Maroulas
- ▶ six paywalled titles recorded as “not obtainable, do not search”

The idea of the runs (1–3 Aug 2026)

- ▶ **I am on vacation.** A computer in the office works alone for one week, running `claude -p` every four hours.
- ▶ **No run has a memory.** The state lives only in `RESEARCH_LOG.md` and in the commits. Every run reads the whole log and appends an entry.
- ▶ **The name says it all.** A ratchet turns in one direction only: nothing verified is ever given up.

A log entry

Read: the current state

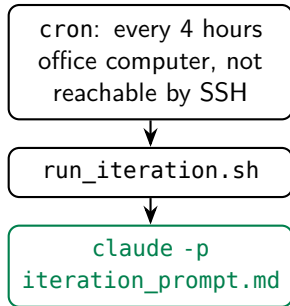
Tried: the approach

Result: PROVED / CONJECTURE / DEAD
END

Next step: one concrete proposal

- ▶ *Honesty before progress.*
- ▶ Dead ends are results.
- ▶ Do not invent literature.
- ▶ Do not retry a dead end without a new reason.
- ▶ No questions back: decide yourself and document it.

The runner: `scripts/run_iteration.sh`



1. **Lock** (flock): runs never overlap.
2. **Set PATH**: otherwise cron cannot find `claude`.
3. Write `STATUS.md`, commit and push: the heartbeat.
4. `claude -p` with a **hard time limit** (90, later 180 minutes) and a **narrow tool list**: `python3`, `git`, `pdftotext`, `ls`, ...
5. Commit and push **even on failure**.
6. Keep the logs of the last 60 runs.

Anything not allowed is refused, not asked about, so a run never waits for me. I switched off system suspend on the machine.

Misstep 1: the run would never have started

The trial run on the morning of 3 Aug found **four errors**, each of which would have killed the unattended run at once:

- ▶ The repository was not at the path the script expected: `cd` fails, the script exits, no log.
- ▶ An `--add-dir` pointing to a missing directory: `claude` aborts immediately.
- ▶ The GPU simulation was somewhere other than stated.
- ▶ An unnecessary `venv` instruction.

More important for the content: `Papers/ratchet/` held far more than `PROBLEM.md` knew about, including `ldp.tex` with the Hamiltonian already derived. Without an entry, the run would have rebuilt it. Since then every file is listed in the problem statement, with a checked “file $\mapsto$ paper” assignment.

Misstep 2: a numerics machine

How it was set up

- ▶ PROBLEM.md: the simulation as “verifier (binding)”
- ▶ prompt: “check numerically”
- ▶ three kinds of result, all ranked equally

Iteration 1 produced four good measurements, no attempt at a proof, and proposed something numerical again.

Supervisor's note, 3 Aug, added to the log by hand, *not* an iteration:

- ▶ **The goal is a theorem, not a table.**
- ▶ The simulation is a **sanity check**: it can refute, never confirm. Anything only supported numerically stays a CONJECTURE.
- ▶ V_* is an infimum, so bounds need **no** LDP:

B1: every path φ gives $V_* \leq I(\varphi)$.

B2: every subsolution F gives
 $V_* \geq \inf\{F : x_0 = 0\}.$

Iteration 2 then proved a negative result: the (x_0, m_1) closure gives no lower bound.

Iterations 1–39: bounds on V_* (3–9 Aug)

- ▶ Up to six runs a day, each ending with a theorem, a negative result or a justified dead end.
- ▶ Examples: iteration 5, “no moment closure of finite order gives a lower bound”; iteration 6, the entropy approach refuted; iteration 9, **the first proved positive lower bound** for $\theta > 1$.
- ▶ State after 39 iterations: V_* bracketed to within $\pm 20\%$ for $\theta \lesssim 2$. This became `paper/quasipotential.tex`.

Misstep 3: the book nobody opened

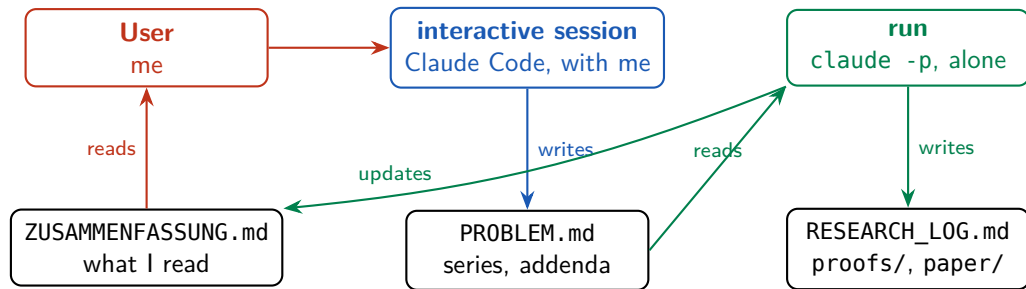
Iteration 39 no longer improved a number at any θ . A search of the log found **zero** hits for “Dawson”, “Kurtz”, “comparison principle”, “tightness”. Dawson & Feng had been available locally from the start. This followed from the setup: the problem statement deliberately bypassed the LDP. The runs work through what is written down, and do not ask whether it is still the right thing.

The change of course: tailored series (from 12 Aug)

Series	Iterations	Dates	Goal	Result
B1–B3	1–39	3–9 Aug	bounds on V_*	bracketed, $\pm 20\%$
B4	40–46	12–14 Aug	the LDP itself	path LDP on a fixed horizon (Thm. 46.4)
B5	47–52	14–15 Aug	the actual click, $\varepsilon \rightarrow 0$	$P(\text{click before } T) = e^{-n(v_T + o(1))}$
B6	53–54	17 Aug	the time horizon	$\lim_T v_T = V_*$ (Thm. 53.4)
B7	55–74	18–23 Aug	the time scale	first click time on the scale e^{nV_*} ; click-rate band
B8	61	18 Aug	moderate deviations	negative after one run, closed

- ▶ Each series has a section in PROBLEM.md with a fixed scope; for B4, **one method per run**. Each iteration delivers a self-contained theorem, “not a quarter four times over”.
- ▶ RATCHET_STOP_AFTER: the series ends by itself, so no run continues without a tailored task.
- ▶ After each iteration an addendum “task from it. N on”. This sets the named goal of the next run.

Three parties, and where the steering happens



- ▶ **Supervisor's notes** in the log, explicitly “not an iteration”: they override the proposed next step.
- ▶ Since 12 Aug every iteration updates **ZUSAMMENFASSUNG.md** (the summary), about ten pages: the only thing a human reads. **FORTSCHRITT.pdf** ended up at 840 pages.

Missteps in operation

Date	Failure	afterwards
12 Aug	The stop limit also counted failed runs: a run dying at the usage limit used up a slot of the series	only <code>rc=0</code> counts
12 Aug	Page images from a 414-page book eat the quota	pdftotext by default
15–20 Aug	Of 27 slots, 21 died at the limit, 19 of them at “Fable 5 limit”, without computing for a single second. <code>--fallback-model</code> does not help with that	runner switched to Opus 5
20 Aug	“session limit” not recognised and reported as a repository error	pattern added
20 Aug	A run stopped by the time limit left no trace: “nothing gets lost” was wrong	first commit within 20 minutes, then one per step
21 Aug	A one-off crontab entry for 21 Aug would have fired again in 2027	removed by hand
23 Aug	Iteration 73 did not write the addendum with the next task	added by hand

87 runs: 63 regular, 24 with an error, almost all at the quota. 74 iterations in terms of content, some of them started by hand.

Missteps in the mathematics

Refuted and documented: this is the normal case

- ▶ The entropy approach to B2 (it. 6): it breaks at the threshold $\theta_c = 2.4088$.
- ▶ (R59-a) refuted and repaired (20 Aug): the product $x_0 m_1$ instead of the pair (x_0, m_1) .
- ▶ The clause (K1) refuted (it. 72), followed by two new routes.
- ▶ B8 (moderate deviations): negative after one run. $c_{MD} \sim 1/(2 \log \theta) \rightarrow 0$ instead of a value in $(0, 1)$. The series had been planned for six runs.

Found only when proofreading by hand (22 Aug, five findings, no statement falls)

- ▶ The introduction claimed a full limit. It is proved only under the hypothesis (RET).
- ▶ A comparison function solved the wrong Riccati equation: the difference is exactly $\theta/2$.
- ▶ A constant cited with $\delta/4$ instead of $\delta/8$. One bound was *smaller* than the one proved.

The runs check their own proofs, but check the **chains of citations** between iterations poorly.

What came out (as of 25 Aug)

Manuscript	Pages	Content
ldp.tex	75	path LDP on a fixed horizon; the actual click; $v_T \downarrow V_*$; first click time on the scale e^{nV_*}
clickrate.tex	16	$\liminf_n \frac{1}{n} \log \liminf_t K^*(t)/t \geq -V_*$, unconditionally and a.s.; the upper half under (RET)
quasipotential.tex	22	B1–B3: bounds on V_* without an LDP
rate.tex	6	the one-dimensional reduction (it. 52)

- ▶ Planned for one week, it ran for three (3–23 Aug).
- ▶ Open: (RET), the existence of τ , the path LDP in D_∞ via Dawson–Gärtner.
- ▶ paper/LEAN_TODO.md (18 Aug): ldp.tex cannot be formalised today; parts of quasipotential.tex can.

What I learned

- ▶ **The problem statement is the lever.** Both big missteps, the numerics machine and the unopened book, were written into PROBLEM.md. The runs did exactly what it said.
- ▶ **Direction stays with the user.** A run does not notice that a route is exhausted. It only refines it.
- ▶ **Tailoring beats freedom.** Since the series, each run has a named goal and a stop limit, and within two weeks the LDP became a manuscript.
- ▶ **A verifier that can only refute is not enough.** The proofs were written in prose. What actually carries them only showed up in proofreading.
- ▶ **Operations cost more than expected.** Paths, quotas, stop counters and leftover crontab entries: each of these failures happened once and has been a comment in the script ever since.